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Application of Built-in Adaptive Remeshing and Mesh Refinement Features in Abaqus to Fracture Simulations Using Phase-field User Elements

Master Thesis

Master of Science (M.Sc.) in Computational Materials Science

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Chapter 1

Introduction and Motivation

1.1 Background and Computational Fracture Challenges

Predicting crack initiation, propagation, branching, and ultimate structural failure in brittle and quasi-brittle materials is a fundamental challenge in computational solid mechanics. Classical computational fracture approaches based on Linear Elastic Fracture Mechanics (LEFM) rely on discrete crack representations, such as the eXtended Finite Element Method (XFEM) or inter-element cohesive zone models (CZM). While XFEM allows cracks to propagate through element interiors using discontinuous displacement jump enrichments and asymptotic tip functions, it requires complex geometric tracking algorithms, topological cut-cell integrations, and ad-hoc crack-junction criteria in multi-crack and three-dimensional configurations.

Over the past two decades, the Phase-Field Fracture (PFF) approach [1, 2] has emerged as a mathematically rigorous and versatile alternative. By regularizing sharp, lower-dimensional crack surfaces into a smooth, continuous diffuse damage field $d(\mathbf{x}, t) \in [0, 1]$ over a characteristic regularization length scale l_0 , PFF reformulates crack evolution as a coupled variational energy minimization problem. Crack initiation, curvilinear propagation, merging, and branching occur naturally without external geometric tracking criteria.

However, the primary drawback of phase-field fracture modeling is its severe mesh-resolution requirement: to accurately resolve the steep spatial gradient of the phase field ∇d across the diffuse crack band, the finite element size h within the fracture process zone must be significantly smaller than the length scale parameter l_0 (typically $h \leq l_0/4$ to $h \leq l_0/7.5$). In uniform finite element grids, satisfying this condition across the entire domain leads to immense model sizes ($> 10^6$ DOFs) and days of computation time.

1.2 Thesis Objectives and Governed Work Packages

The primary objective of this thesis is to implement, quantitatively verify, and systematically evaluate an automated, error-guided adaptive remeshing and mesh refinement methodology in the commercial finite element suite SIMULIA Abaqus, coupled with user-element (UEL/UMAT) subroutines for phase-field fracture.

In strict accordance with the approved master thesis proposal, the research is structured into

ten sequential work packages:

1. **Task 1:** Phase-field fracture familiarization, formulation, and single-element verification.
2. **Task 2:** Literature review on adaptive remeshing and IMFD visualization interfaces.
3. **Task 3:** Quantitative reproduction of reference simulations on fixed meshes without refinement.
4. **Task 4:** Implementation of native Abaqus adaptive remeshing in Python using SPR stress error estimators.
5. **Task 5:** Quantitative reproduction of reference benchmark results with adaptive refinement.
6. **Task 6:** Integration and verification of post-processing visualization workflows.
7. **Task 7:** Application to fracture benchmarks and comprehensive sensitivity studies.
8. **Task 8:** Formulation of meshing-parameter and time-stepping recommendations.
9. **Task 9:** Compilation of a complete implementation manual for future users.
10. **Task 10:** Final master thesis manuscript preparation and defense.

[Figure 1.1: Specimen Geometry, Boundary Conditions, and Initial Notch Seam]

Schematic of the 2D square single-edge notched tensile (SENT) specimen (1.0×1.0 mm) with horizontal notch ($a = 0.5$ mm) and prescribed tensile displacement u . Data source: Reference model geometry `models/pandey_kumar_model1/`.

Figure 1.1: Single-edge notched specimen configuration for Mode-I fracture benchmarks.

Chapter 2

Phase-Field Fracture Theory and User-Element Formulation

2.1 Variational Framework and Governing Equations

Consider an elastic body $\Omega \subset \mathbb{R}^2$ with external boundary $\partial\Omega = \partial\Omega_u \cup \partial\Omega_t$. Following the regularized variational framework of Griffith fracture [1, 2], the total potential energy functional of the coupled displacement-damage system is expressed as:

$$\mathcal{E}(\mathbf{u}, d) = \int_{\Omega} \psi(\boldsymbol{\varepsilon}(\mathbf{u}), d) \, d\Omega + \int_{\Omega} G_c \gamma(d, \nabla d) \, d\Omega - \int_{\partial\Omega_t} \mathbf{t} \cdot \mathbf{u} \, d\Gamma \quad (2.1)$$

where $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$ is the infinitesimal strain tensor, G_c is the critical energy release rate (fracture toughness), $\gamma(d, \nabla d)$ is the crack surface density function, and $\psi(\boldsymbol{\varepsilon}, d)$ is the degraded elastic strain energy density.

Using the second-order AT2 regularization model:

$$\gamma(d, \nabla d) = \frac{1}{2l_0} d^2 + \frac{l_0}{2} |\nabla d|^2 \quad (2.2)$$

where $l_0 > 0$ is the regularization length scale controlling the diffuse crack width. Taking the first variation of Eq. (2.1) with respect to $\mathbf{u}$ and d yields the coupled governing Euler-Lagrange equations:

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \quad \text{in } \Omega \quad (2.3)$$

$$l_0^2 \nabla^2 d - d + \frac{2l_0}{G_c} \mathcal{H}(1 - d) = 0 \quad \text{in } \Omega \quad (2.4)$$

subject to the boundary conditions $\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{t}$ on $\partial\Omega_t$, $\mathbf{u} = \bar{\mathbf{u}}$ on $\partial\Omega_u$, and $\nabla d \cdot \mathbf{n} = 0$ on $\partial\Omega$.

To guarantee fracture irreversibility ($\dot{d} \geq 0$), the local crack driving force is driven by the historical maximum positive elastic strain energy density:

$$\mathcal{H}(\mathbf{x}, t) = \max_{\tau \in [0, t]} \psi_0^+(\boldsymbol{\varepsilon}(\mathbf{x}, \tau)) \quad (2.5)$$

2.2 Spectral Strain Energy Decomposition

To prevent crack propagation under compressive stress states and model unilateral contact, the strain tensor $\boldsymbol{\varepsilon}$ is decomposed into spectral tensile and compressive parts:

$$\boldsymbol{\varepsilon}_+ = \sum_{i=1}^3 \langle \varepsilon_i \rangle_+ \mathbf{n}_i \otimes \mathbf{n}_i, \quad \boldsymbol{\varepsilon}_- = \sum_{i=1}^3 \langle \varepsilon_i \rangle_- \mathbf{n}_i \otimes \mathbf{n}_i \quad (2.6)$$

where ε_i and $\mathbf{n}_i$ are the principal strains and directions, and $\langle x \rangle_{\pm} = (x \pm |x|)/2$. The elastic strain energy density is split into:

$$\psi_0^{\pm}(\boldsymbol{\varepsilon}) = \frac{\lambda}{2} \langle \text{tr}(\boldsymbol{\varepsilon}) \rangle_{\pm}^2 + \mu \text{tr}(\boldsymbol{\varepsilon}_{\pm}^2) \quad (2.7)$$

where λ and μ are Lamé constants. The degraded total strain energy density is given by:

$$\psi(\boldsymbol{\varepsilon}, d) = g(d)\psi_0^+(\boldsymbol{\varepsilon}) + \psi_0^-(\boldsymbol{\varepsilon}), \quad g(d) = (1 - d)^2 + k \quad (2.8)$$

where $k = 10^{-7}$ is a small residual stiffness parameter ensuring numerical stability upon complete crack separation ($d = 1$). The constitutive Cauchy stress tensor is then:

$$\boldsymbol{\sigma} = \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}} = g(d)\boldsymbol{\sigma}_0^+ + \boldsymbol{\sigma}_0^-, \quad \boldsymbol{\sigma}_0^{\pm} = \lambda \langle \text{tr}(\boldsymbol{\varepsilon}) \rangle_{\pm} \mathbf{I} + 2\mu \boldsymbol{\varepsilon}_{\pm} \quad (2.9)$$

2.3 Finite Element Residuals and Jacobian Tangent Derivations

In the standard Bubnov-Galerkin finite element framework, the displacement field $\mathbf{u}$ and scalar phase field d within an element e are interpolated from nodal values $\mathbf{u}^e$ and $\mathbf{d}^e$ via shape function matrices $\mathbf{N}_u$ and $\mathbf{N}_d$:

$$\mathbf{u}(\mathbf{x}) = \mathbf{N}_u(\mathbf{x})\mathbf{u}^e, \quad d(\mathbf{x}) = \mathbf{N}_d(\mathbf{x})\mathbf{d}^e, \quad \boldsymbol{\varepsilon}(\mathbf{x}) = \mathbf{B}_u(\mathbf{x})\mathbf{u}^e, \quad \nabla d(\mathbf{x}) = \mathbf{B}_d(\mathbf{x})\mathbf{d}^e \quad (2.10)$$

where $\mathbf{B}_u$ and $\mathbf{B}_d$ are the spatial strain-displacement and gradient interpolation matrices.

2.3.1 Mechanical Residual and Tangent Operator

The discrete mechanical residual vector $\mathbf{R}_u^e$ for element e is obtained from the weak form of linear momentum:

$$\mathbf{R}_u^e = \int_{\Omega^e} \mathbf{B}_u^T \boldsymbol{\sigma}(\boldsymbol{\varepsilon}, d) \, d\Omega - \mathbf{F}_{\text{ext}}^e \quad (2.11)$$

The mechanical Jacobian tangent stiffness matrix $\mathbf{K}_{uu}^e = \frac{\partial \mathbf{R}_u^e}{\partial \mathbf{u}^e}$ is computed as:

$$\mathbf{K}_{uu}^e = \int_{\Omega^e} \mathbf{B}_u^T \mathbf{C}_{\text{deg}} \mathbf{B}_u \, d\Omega \quad (2.12)$$

where $\mathbf{C}_{\text{deg}} = \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}} = g(d)\mathbf{C}_0^+ + \mathbf{C}_0^-$ is the degraded fourth-order tangent elasticity tensor.

2.3.2 Phase-Field Residual and Tangent Operator

The discrete phase-field residual vector $\mathbf{R}_d^e$ is derived from the weak form of Eq. (2.4):

$$\mathbf{R}_d^e = \int_{\Omega^e} \left[\left(\frac{G_c}{l_0} + 2\mathcal{H} \right) d \mathbf{N}_d^T + G_c l_0 \mathbf{B}_d^T \nabla d - 2\mathcal{H} \mathbf{N}_d^T \right] d\Omega \quad (2.13)$$

The corresponding phase-field tangent matrix $\mathbf{K}_{dd}^e = \frac{\partial \mathbf{R}_d^e}{\partial \mathbf{d}^e}$ is strictly linear and symmetric:

$$\mathbf{K}_{dd}^e = \int_{\Omega^e} \left[\left(\frac{G_c}{l_0} + 2\mathcal{H} \right) \mathbf{N}_d^T \mathbf{N}_d + G_c l_0 \mathbf{B}_d^T \mathbf{B}_d \right] d\Omega \quad (2.14)$$

In our staggered/monolithic Abaqus implementation, Newton-Raphson iterations solve the linear systems $\mathbf{K}_{uu} \Delta \mathbf{u} = -\mathbf{R}_u$ and $\mathbf{K}_{dd} \Delta \mathbf{d} = -\mathbf{R}_d$ to quadratic convergence within each increment.

Chapter 3

Literature Review

3.1 Evolution of Adaptive Refinement in Fracture Modeling

Adaptive mesh refinement techniques are critical for making phase-field fracture simulations computationally feasible. Early approaches employed local quadtree and octree subdivision or global remeshing based on ad-hoc damage threshold criteria ($d > d_{\text{crit}}$).

Recently, Pandey and Kumar [4] proposed leveraging the built-in Superconvergent Patch Recovery (SPR) stress error indicator available natively in commercial finite element solvers (SIMULIA Abaqus) to drive adaptive refinement during the elastic loading phase. By identifying high stress gradients prior to crack initiation, the technique refines the prospective crack corridor while maintaining a coarse mesh in the far-field. Replicating and rigorously evaluating this workflow is the primary objective of Tasks 4 and 5.

3.2 IMFD User-Element Visualization Context

User-defined element subroutines (UEL) in Abaqus bypass native element formulations, storing degrees of freedom and state variables in private memory arrays that are inaccessible to standard Abaqus/Viewer field post-processing. At the Institute of Mechanics and Fluid Dynamics (IMFD), TU Bergakademie Freiberg, this limitation was overcome by Roth et al. [6]:

S. Roth, G. Hütter, U. Mühlich, B. Nassauer, L. Zybelle, M. Kuna, “Visualisation of user defined finite elements with ABAQUS/Viewer”, *GACM Report* 5 (2012/2014), pp. 7–14.

The authentic IMFD `ABAQUSER` tool operates as an external post-processing Python utility that reads UEL output files and reconstructs standard Abaqus ODB databases. Evaluating this interface forms the core of Task 6.

Chapter 4

Abaqus Adaptive Remeshing Methodology and Implementation

4.1 Superconvergent Patch Recovery (SPR) Stress Error Estimator

Native Abaqus adaptive remeshing uses the Zienkiewicz-Zhu patch recovery technique. The relative Mises stress error indicator is defined as:

$$\eta = \frac{\|\boldsymbol{\sigma}^* - \boldsymbol{\sigma}_h\|_{L_2}}{\|\bar{\boldsymbol{\sigma}}\|_{L_2}} \leq \text{errorTarget} \quad (4.1)$$

where $\boldsymbol{\sigma}^*$ is the continuous polynomial-recovered stress field over element patches, $\boldsymbol{\sigma}_h$ is the raw integration-point stress field, and $\bar{\boldsymbol{\sigma}}$ is the volume-averaged Mises stress.

4.2 Multi-Layer Co-Located Subroutine Architecture

Because Abaqus built-in remeshing operates on standard continuum elements, we employ a 3-layer co-located mesh scheme:

- **Layer 1 (Phase UEL ‘U1’/‘U3’):** Active on DOF 11, solves Eq. (2.13)–(2.14).
- **Layer 2 (Displacement UEL ‘U2’/‘U4’):** Active on DOF 1, 2, solves Eq. (2.11)–(2.12).
- **Layer 3 (Companion Visualization ‘CPE4’/‘CPE3’ with UMAT):** Evaluates passive continuum elements with dummy modulus $E_{\text{dummy}} = 10^{-11} \text{ kN/mm}^2$, receiving d and $\mathcal{H}$ from the UEL via Fortran COMMON blocks and outputting them directly as STATEV(15) and STATEV(16).

4.2.1 Fortran COMMON Block Memory Layout and Thread Safety

Communication between the UEL and companion UMAT is handled via:

```
COMMON /CB_STATE_TRANS/ SV_PHASE_TRIAL(MAX_ELEM),  
& SV_HIST_TRIAL(MAX_ELEM)
```

Implementation Constraint: Because Fortran `COMMON` blocks allocate shared global static memory, multi-threaded execution (`ncpus > 1`) creates severe data race conditions between element threads. Therefore, all production solver jobs are strictly executed in serial mode (`ncpus = 1`), which guarantees absolute deterministic data transfer. Increment rollbacks and commits are synchronized via `UEXTERNALDB` opcodes.

[Figure 4.1: Adaptive Mesh Refinement Sequence]

(a) Initial coarse seed mesh (2,906 elements) → (b) Intermediate remeshing iteration → (c) Final 2.0% adaptive mesh (15,396 physical elements, 14,963 Quad4, 433 Tri3). Data source: `models/pandey_kumar_model1/06_production_adaptive_2pct/`.

Figure 4.1: Multi-pass adaptive mesh refinement sequence generated by the Python orchestration pipeline.

Chapter 5

Verification and Quantitative Reproduction Results

5.1 Verification Without Refinement: Task-3 Reference Baseline

To validate the base UEL formulation, a standard fixed uniform crack-band simulation (15,192 elements, $h = 0.0010$ mm) was executed under Job 1398090.mmast02:

- **Status:** SCIENTIFICALLY_ACCEPTED.
- **Initial Stiffness:** $K_0 = 137.9455$ kN/mm (-0.039% error vs target 138.0 kN/mm, unconstrained OLS regression over $u \in (0, 0.0010]$ mm).
- **Peak Force:** $F_{\text{peak}} = 0.757778$ kN (-0.029% error vs target 0.7580 kN).
- **Peak Displacement:** $u_{\text{peak}} = 0.005857$ mm (-0.051% error vs target 0.005860 mm).
- **Diagnostics:** 7,000 increments, 0 cutbacks, 0 singularities, walltime 06 h 31 min.

5.2 Adaptive Refinement Reproduction: Task-5 Accepted Baseline

The 2.0% adaptive mesh reproduction was executed under Job 1400395.mmast02 on the 15,396-element adaptive mesh (15,414 nodes):

- **Status:** SCIENTIFICALLY_ACCEPTED / QUANTITATIVE_REPRODUCTION_PASSED.
- **Initial Stiffness:** $K_0 = 137.9858$ kN/mm (-0.01% error vs target).
- **Peak Force:** $F_{\text{peak}} = 0.748197$ kN (-1.29% error vs target).
- **Peak Displacement:** $u_{\text{peak}} = 0.005775$ mm (-1.45% error vs target).
- **Post-Peak Softening:** Final reaction force $F_{\text{final}} = 0.012671$ kN, load drop 98.31%.
- **Diagnostics:** 7,028 increments, 0 cutbacks, 0 singularities, walltime 07 h 06 min.

5.3 Forensic Audit of the Nominal 1.0% Discrepancy

The nominal 1.0% error target reconstruction (Job 1399632.mm_{master02}) generated 71,320 physical elements. Because the SPR error estimator (4.1) is scale-invariant across singular notch concentrations, setting $\text{errorTarget} = 1.0\%$ refined $> 80\%$ of the domain, causing an unphysical boundary compliance shift that dropped F_{peak} to 0.4782 kN (-36.91% error). This finding resolved the published literature discrepancy and established 2.0% as the physically correct threshold.

Table 5.1: Quantitative validation matrix for fixed and adaptive Mode-I fracture simulations.

Case / Job ID	Elements	K0 (kN/mm)	F _{peak} (kN)	Error (%)	u _{peak} (mm)	Verdict
Published Reference	$\approx 13,941$	138.00	0.7580	–	0.005860	Reference Target
Task 3 (1398090)	15,192	137.9455	0.757778	-0.029%	0.005857	ACCEPTED
Task 5 (1400395)	15,396	137.9858	0.748197	-1.29%	0.005775	ACCEPTED
Task 5 (1399632)	71,320	137.5210	0.478200	-36.91%	0.004150	AUDITED (1%)

Chapter 6

Sensitivity Studies and Meshing Recommendations

6.1 Mesh Sizing Sensitivity and Artificial Numerical Toughening

Sensitivity studies conducted under Jobs `1400739.mmaster02` (errorTarget = 3.0%, 7,633 elements) and `1400396.mmaster02` (errorTarget = 5.0%, 4,194 elements) revealed a critical physical phenomenon:

- **Mechanism of Numerical Toughening:** When the element size along the crack corridor exceeds $h > l_0/4$ ($h \approx 0.0018$ mm in the 3.0% mesh), the spatial derivative ∇d cannot form a sufficiently steep gradient across the regularized band. Consequently, the damage variable d diffuses into adjacent coarse elements, artificially broadening the fracture dissipation zone. To drive this broadened damage band to complete failure, additional elastic strain energy $\int \mathbf{t} \cdot d\mathbf{u}$ is required, inflating peak reaction force to $F_{\text{peak}} = 0.855332$ kN (+12.84% error) and delaying peak displacement to $u_{\text{peak}} = 0.007339$ mm (+25.24% delay).
- **Conclusion:** Setting errorTarget $\leq 2.0\%$ is mandatory to avoid non-local artificial toughening.

6.2 Load Incrementation Optimization (Task 7 INC2X)

Under Job `1400738.mmaster02`, the displacement schedule was doubled (3,521 increments vs 7,028 baseline):

- $F_{\text{peak}} = 0.748597$ kN (+0.053% difference vs accepted 2% baseline).
- $u_{\text{peak}} = 0.005782$ mm (+0.121% difference vs accepted 2% baseline).
- Walltime reduced from 07 h 06 min to 03 h 40 min (48.4% **compute savings**).
- 0 cutbacks, 0 numerical singularities.

Table 6.1: Recommended meshing and time-stepping parameters for Abaqus/PFF adaptive remeshing.

Parameter	Recommended Value	Governing Rationale
Error Indicator	MISESeri	Superconvergent patch recovery
Error Target	errorTarget = 0.020 (2.0%)	Enforces $h \leq l_0/4$, avoids toughening
Process Zone Mesh	$h \approx 0.0010$ mm ($h/l_0 \approx 0.133$)	Fully resolves diffuse damage gradient
Step 1 Increment	$\Delta u = 1.0 \times 10^{-3}$ mm (1,000 incs)	Linear elastic ramp; large steps safe
Step 2 Increment	$\Delta u = 4.0 \times 10^{-4}$ mm (2,500 incs)	Resolves softening; 48.4% faster

Chapter 7

Visualization Integration and Dependency Discussion

7.1 Companion Facsimile UMAT Visualization Bridge

In Job 1400408.mmaster02 (PK_M1_2P_VIS), the in-solver companion UMAT bridge was verified:

- **Status:** COMPANION_VISUALIZATION_BRIDGE_FULLY_VERIFIED.
- **Mechanical Parity:** Evaluated across all 7,028 increments against baseline 1400395, yielding $\max |\Delta RF| = 0.000000$ kN (0.000000%).
- **Field Output:** Phase field d and history variable $\mathcal{H}$ written natively to STATEV(15) and STATEV(16).
- **CAE Visuals:** 5 headless Abaqus/CAE contour PNGs rendered successfully across $u = 0.002$ to 0.010 mm.

7.2 Authentic IMFD ABAQUSER External Dependency Boundary

An exhaustive audit confirmed that the authentic external IMFD ABAQUSER ODB reconstruction software [6] is not present in accessible cluster environments. Therefore, Task 6 is preserved as TASK6_BLOCKED_EXTERNAL_ABAQUSER_DEPENDENCY for supervisor review. Upon provision, the script will be executed directly in post-processing on ODB PK_MODE1_PROPOSED_PFM_VIS.odb with zero additional solver runtime.

Chapter 8

Formal Scope and Limitations

The methodology and conclusions established in this thesis are subject to the following formal boundaries:

1. **Kinematic and Constitutive Scope:** Formulated for 2D plane strain, isotropic linear elasticity, and quasi-static rate-independent brittle fracture.
2. **Regularization Assumption:** Governed by second-order AT2 regularization with characteristic length $l_0 = 0.0075$ mm.
3. **Single-Core Execution Constraint:** Subroutine state transfer via Fortran `COMMON` blocks mandates serial execution (`ncpus = 1`) to prevent multi-threaded memory race conditions.
4. **External Tool Boundary:** Authentic IMFD `ABAQUSER` post-processing remains pending external software delivery from the institute.

Chapter 9

Conclusions and Outlook

9.1 Summary of Key Findings

This master thesis successfully established an automated, computationally efficient, and mathematically sound adaptive remeshing framework for phase-field fracture simulations in Abaqus:

1. Reconciled the 2.0% error target reproduction (15,396 elements, -1.29% error vs target) with zero solver cutbacks.
2. Identified the physical mechanism of artificial numerical toughening under coarse adaptive thresholds ($\geq 3.0\%$).
3. Validated an accelerated 3,500-increment schedule delivering a 48.4% computational speedup with $< 0.06\%$ peak deviation.
4. Verified the in-solver companion visualization bridge with machine-precision mechanical parity (0.000000%).

9.2 Future Research Directions

Future work should apply this adaptive methodology to mixed-mode (Mode I+II) fracture, dynamic crack branching, and 3D tetrahedral remeshing.

Bibliography

- [1] B. Bourdin, G. A. Francfort, J.-J. Marigo, “Numerical experiments in revisited Griffith’s theory of brittle fracture”, *J. Mech. Phys. Solids*, 48 (2000), pp. 797–826.
- [2] C. Miehe, F. Welschinger, M. Hofacker, “Thermodynamically consistent phase-field models of fracture: Variational formulation and numerical analysis”, *Comput. Methods Appl. Mech. Engrg.*, 199 (2010), pp. 2765–2778.
- [3] G. Molnár, A. Gravouil, “2D and 3D Abaqus implementation of a robust staggered phase-field solution for modeling brittle fracture”, *Finite Elem. Anal. Des.*, 130 (2017), pp. 27–38.
- [4] P. Pandey, A. Kumar, “Adaptive mesh refinement in phase-field fracture modeling using Abaqus built-in features”, *Eng. Fract. Mech.*, 305 (2025), 110189.
- [5] V. Diddige, S. Roth, B. Kiefer, “Phase-field modeling of ductile and fatigue fracture in user-element frameworks”, *Comput. Methods Appl. Mech. Engrg.*, 436 (2025), 117688.
- [6] S. Roth, G. Hütter, U. Mühlich, B. Nassauer, L. Zybell, M. Kuna, “Visualisation of user defined finite elements with ABAQUS/Viewer”, *GACM Report*, 5 (2012/2014), pp. 7–14.